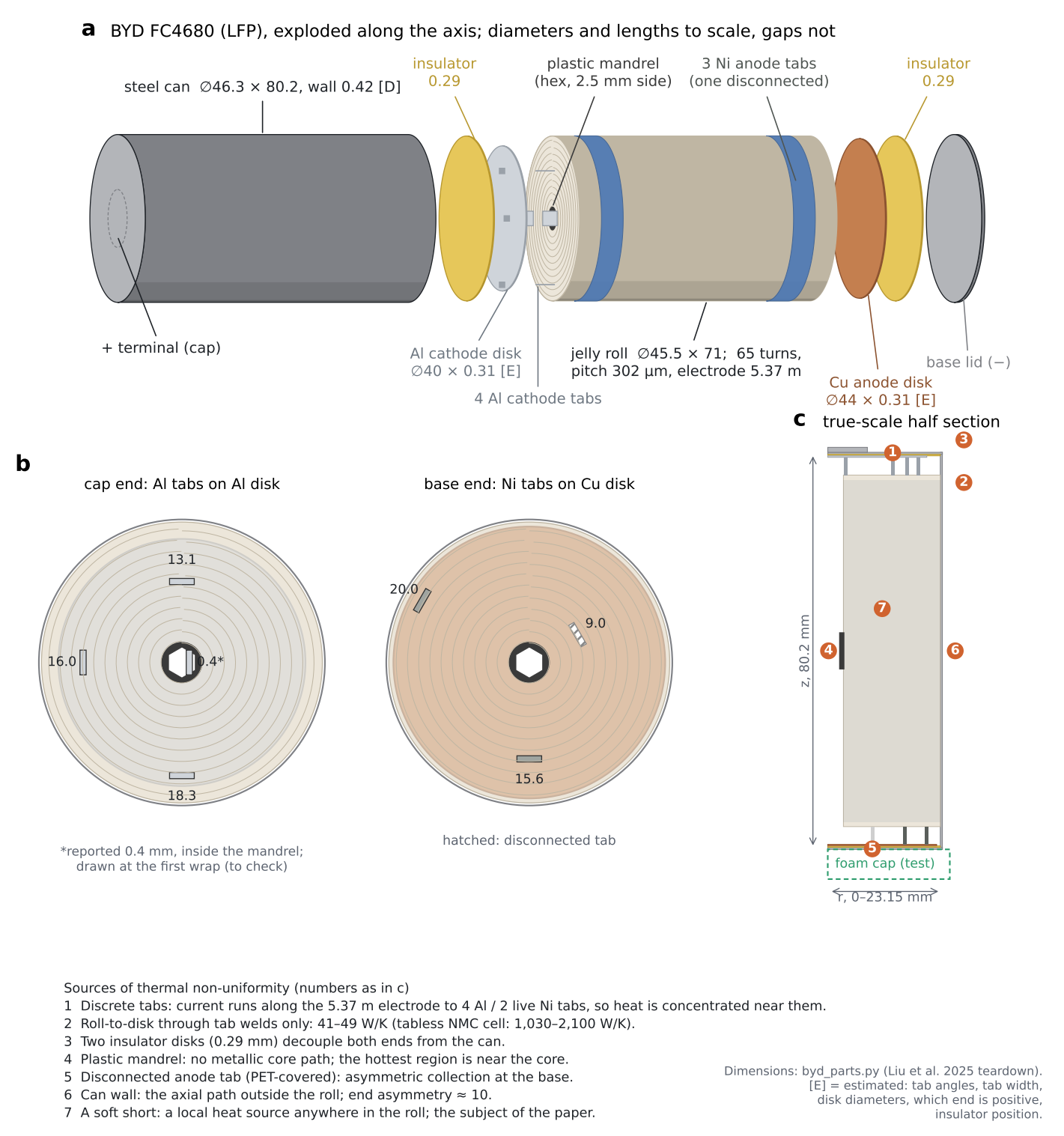


Sources of thermal non-uniformity (numbers as in c)

1 Discrete tabs: current runs along the 5.37 m electrode to 4 Al / 2 live Ni tabs, so heat is concentrated near them.
2 Roll-to-disk through tab welds only: 41–49 W/K (tabless NMC cell: 1,030–2,100 W/K).
3 Two insulator disks (0.29 mm) decouple both ends from the can.
4 Plastic mandrel: no metallic core path; the hottest region is near the core.
5 Disconnected anode tab (PET-covered): asymmetric collection at the base.
6 Can wall: the axial path outside the roll; end asymmetry ≈ 10.
7 A soft short: a local heat source anywhere in the roll; the subject of the paper.

Dimensions: byd_parts.py (Liu et al. 2025 teardown). [E] = estimated: tab angles, tab width, disk diameters, which end is positive, insulator position.